MILITARY SPECIFICATION SHEET
GUIDANCE CONTROL SECTION, GUIDED MISSILE
SEEKER SECTION ASSEMBLY

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the seeker section assembly described herein shall consist of this specification and the latest issue of MIL-G-85742.

1. SCOPE

1.1 Scope. This specification sheet establishes the performance and test requirements for the seeker section assembly used in the Guidance Control Section (GCS) of the SIDEARM Missile.

2. APPLICABLE DOCUMENTS

2.1 Government drawings. The following drawings, in addition to the applicable documents listed in MIL-G-85742, are applicable to this specification sheet.

Naval Air Systems Command
(Code Ident 30003)

1466AS1065	Gyro-Rotor Assembly.
1466AS1066	Antenna Assembly.
1466AS1082	Seeker Section Assembly.
1466AS1338	Cable Assembly Special Purpose, Electrical, Branched.

FSC 1420

AMSC N/A

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3. REQUIREMENTS

3.1 Item description. The seeker section assembly is comprised of the following:

a. Gyro-Rotor Assembly (Drawing 1466AS1065).

b. Antenna Assembly (Drawing 1466AS1066).

c. Cable Assembly (Drawing 1466AS1338).

3.2 Characteristics. Performance characteristics of the seeker section assembly shall be as specified herein.

3.2.1 Antenna.

3.2.1.1 Gain. The gain of the antenna shall be greater than 0 decibels (dB) with respect to a linear isotropic antenna at the peak of the main lobe.

3.2.1.2 Beamwidth. The width of the main lobe 3 dB below the peak shall be less than 20 degrees (deg).

3.2.1.3 Sidelobes. The sidelobes shall be 6 dB below the main lobe.

3.2.1.4 Conical symmetry. The difference between the mean values of diametrically opposite points around boresight shall be less than 2 dB.

3.2.2 Gyro assembly.

3.2.2.1 Gyro drive voltage. The seeker three-phase Y connected gyro drive voltage shall be 19.0 ±1.5 volts root-mean-square, line-to-neutral, at a frequency of 320 ±5 hertz (Hz).

3.2.2.2 Gyro steady-state speed. The gyro steady-state speed with the gyro caged and the radome in place shall be as specified in MIL-G-85742, Appendix.

3.2.2.3 Gyro rotation. When looking from the front of the antenna the gyro shall rotate clockwise.

3.2.2.4 Gyro slowdown. Ten seconds after removing power to the gyro, the gyro speed shall decrease not less than 7 Hz and no more than 20 Hz from its steady-state value (see 3.2.2.2).

3.2.2.5 Gyro static balance. The gyro rotor shall be statically balanced such that it rests not more than 10 deg off boresight.

3.2.2.6 Gyro dynamic balance. The gyro rotor assembly shall be dynamically balanced to 300 micro-ounce inches or better.

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3.3 Physical characteristics. Physical characteristics of the seeker section assemblies shall be in accordance with the drawings specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. Each seeker section assembly shall meet the requirements of 3.2 when inspected and tested as specified in Drawing 1466AS1082 and the following paragraphs. Failure to meet any of these requirements shall be cause for rejection of the seeker section assembly.

4.1.1 Test environment. Unless otherwise specified, all tests and inspections shall be conducted under the following environmental conditions:

a. Temperature (ambient): 23 ±5 degrees Celsius.

b. Altitude: Elevation of test facility.

c. Relative humidity: 95 percent or less.

4.1.2 Test equipment accuracy. Unless otherwise specified, test equipment accuracy shall be as specified in MIL-G-85742.

4.2 Quality conformance inspection.

4.2.1 Unit test. The assembly shall be subjected to a unit test in order to establish conformance to the requirements specified herein.

4.2.2 Test methods. Details of methods and procedures shall be in accordance with the Government approved test procedure.

4.2.2.1 Antenna gain. The gain shall be measured at frequencies A, D, and G. The measurements shall meet the requirements of 3.2.1.1.

4.2.2.2 Antenna beamwidth and sidelobes. Antenna beamwidth and sidelobe characteristics shall be measured at frequencies A, D, and G. These measurements shall meet the requirements of 3.2.1.2 and 3.2.1.3.

4.2.2.3 Conical symmetry. The conical symmetry shall be measured at frequencies A, D, and G. The measurements shall meet the requirements of 3.2.1.4.

4.2.2.4 Gyro drive voltage. The gyro drive voltage of 3.2.2.1 shall be applied at pins 1, 2, and 3 of the seeker connector in that order for proper phase rotation.

4.2.2.5 Gyro measurements. With the drive voltage of 4.2.2.4 applied, measure the gyro steady-state speed on pin 20 of the seeker connector with pin 8 as ground. The speed and direction of rotation shall meet the requirements of 3.2.2.2 and 3.2.2.3 respectively. Remove the drive voltage and verify that the gyro meets the requirements of 3.2.2.4.

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4.2.2.6 Balance.

4.2.2.6.1 Static balance. The static balance shall be adjusted by modifying weights on the back of the stator. The static balance shall meet the requirements of 3.2.2.5.

4.2.2.6.2 Dynamic balance. The gyro rotor dynamic balance shall be adjusted by removing material from the gyro rotor. It shall meet the requirements of 3.2.2.6.

Preparing activity
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